Unsupervised Analysis of Screen Time and Mental Wellness Behaviours

Unsupervised machine learning analysis (PCA, K-Means, Hierarchical Clustering)

Author: Mohamed Amine Laabidi

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Abstract

This report presents an unsupervised machine learning analysis of 400 participants' screen time habits, lifestyle behaviours, and mental wellness scores. Using Principal Component Analysis (PCA) for dimensionality reduction followed by Hierarchical and K-Means clustering, the analysis identifies three behavioural profiles without using the mental wellness index to build them. The segments differ substantially in stress levels, sleep, and screen time patterns, and show clearly different wellness levels when profiled afterwards. Both clustering methods agree on 84.5% of assignments after label matching, although silhouette scores (0.29–0.34) indicate that the groups overlap rather than being sharply separated.

1. Problem Statement

Predicting mental wellness from screen time data using supervised models reveals which variables matter statistically, but it does not reveal whether participants naturally form distinct behavioural groups. This report addresses the question: do participants cluster into identifiable lifestyle profiles based on their screen time and health behaviours alone, and do those profiles correspond to different wellness outcomes?

2. Data Overview

The dataset is the public Screen Time vs Mental Wellness Survey – 2025 (Kaggle): 400 participant records collected to study the relationship between digital screen habits and mental wellbeing. It contains 15 variables spanning demographics, screen time behaviour, lifestyle factors, and wellbeing outcomes; one binary indicator (sleep_quality_binary) was derived from them.

2.1 Variable Summary

Variable	Type	Description
user_id	ID	Unique participant identifier
age	Numeric	Age in years
gender	Categorical	Self-reported gender
occupation	Categorical	Employed, Self-employed, Student, Unemployed, Retired
work_mode	Categorical	Remote, Hybrid, In-person
screen_time_hours	Numeric	Total daily screen time
work_screen_hours	Numeric	Daily screen time for work purposes
leisure_screen_hours	Numeric	Daily screen time for leisure
sleep_hours	Numeric	Average sleep duration
sleep_quality_1_5	Ordinal	Sleep quality rating (1–5)
sleep_quality_binary	Binary	Derived: 1 = poor sleep
stress_level_0_10	Numeric	Self-reported stress (0–10)
exercise_minutes_per_week	Numeric	Weekly physical activity in minutes
social_hours_per_week	Numeric	Weekly hours of social interaction
productivity_0_100	Numeric	Self-reported productivity score (0–100)
mental_wellness_index_0_100	Numeric	Mental wellness index (0–100), used for profiling only

2.2 Descriptive Statistics (Numeric Variables)

Variable	Min	Mean	Median	Max	Std Dev
age	16	29.78	30.00	60	7.47
stress_level_0_10	0	8.15	8.80	10	2.09
sleep_hours	4.64	7.01	7.03	9.74	0.85

work_screen_hours	0.11	2.18	1.46	12.04	1.93
leisure_screen_hours	0.89	6.84	6.70	13.35	2.22
exercise_min/week	0	109.81	103	372	70.01
social_hours/week	0	7.90	7.75	23.90	4.91
mental_wellness_index	0	20.33	14.80	97	20.38

The mental wellness index is strongly right-skewed: three quarters of participants score below 31, with a mean of 20.3 and a median of 14.8, and 15% score exactly 0. Stress levels are skewed toward the high end, with a mean of 8.15 out of 10 and a third of participants at the maximum of 10.

3. Data Cleaning & Preprocessing

3.1 Variable Removal

Variable Removed
user_id
screen_time_hours
productivity_0_100
sleep_quality_1_5
mental_wellness_index_0_100
gender, occupation, work_mode

3.2 Scaling:

All eight numeric variables were standardised to zero mean and unit variance using z-score scaling. This step is essential for PCA and K-Means, both of which are sensitive to variable scale. Without scaling, variables with larger ranges (e.g., exercise_minutes_per_week, up to 372) would dominate the distance computation.

3.3 Final Dataset for Analysis

- 400 observations
- 8 numeric variables for PCA and clustering
- 3 categorical variables for profiling only
- 1 wellness target retained separately for cluster profiling
- 0 missing values, 0 duplicates

4. Key Findings

1. Three participant profiles: Balanced (44%), Work-Screen Stressed (18%), and Leisure-Screen Stressed (38%). The two stressed groups both have low wellness but differ in the type of screen use.

2. Stress and sleep are the dominant axes of variation: PC1 is defined by stress_level_0_10 (30%), sleep_hours (26%), and sleep_quality_binary (24%), explaining 30.2% of total variance.
3. Screen type matters: both high-stress clusters have high total screen time (11.0 and 9.8 h/day) but different compositions, dominated by work screen time (5.17 h/day) in one and by leisure screen time (8.51 h/day) in the other.
4. The clustering solution is reasonably stable: K-Means and Hierarchical agree on 84.5% of assignments after correcting for label switching. Silhouette scores of 0.29–0.34 indicate weak-to-moderate separation, so the profiles describe tendencies rather than sharply distinct groups.
5. Work screen time separates the two high-stress groups: its correlation with wellness is weak ($r = -0.29$), but it dominates PC2 and is the main variable distinguishing Cluster 1, almost entirely remote workers (98.6%), from Cluster 3.

5. Theoretical Background

5.1 Principal Component Analysis (PCA)

PCA is a linear dimensionality reduction technique that transforms correlated variables into mutually orthogonal principal components, ordered by the share of variance they explain. Each component is a linear combination of the original variables, whose weights are called loadings. Keeping only the first components reduces noise, removes correlation between features, and makes Euclidean distances more meaningful for clustering.

5.2 Hierarchical Clustering (Ward Linkage)

Hierarchical clustering builds a tree (dendrogram) of nested groups by repeatedly merging the two closest clusters, starting from individual observations. Ward's method merges, at each step, the pair of clusters whose union produces the smallest increase in total within-cluster variance, so the resulting clusters are as homogeneous as possible. Large jumps in merge height in the dendrogram suggest a natural number of clusters K; here this is complemented by silhouette scores.

5.3 K-Means Clustering

K-Means partitions n observations into K clusters by iteratively assigning each point to its nearest centroid and updating centroids as the cluster mean, minimising total within-cluster sum of squares. K-Means benefits from PCA pre-processing which removes collinearity.

5.4 Silhouette Score

The silhouette coefficient for observation i is $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$, where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean distance to the nearest neighbouring cluster. Following Kaufman and Rousseeuw (1990), an average silhouette of 0.26–0.50 indicates weak structure, 0.51–0.70 reasonable structure, and above 0.70 strong structure.

6. Analysis

6.1 PCA Results

The first two components retain 46.2% of the total variance and the first three 59.7%. Clustering was performed on the first three components (PC1–PC2 are used for visualisation): from PC4 onwards each component explains about as much variance as a single original variable (eigenvalue ≈ 1 or less), so adding them mostly adds noise to the distance computation.

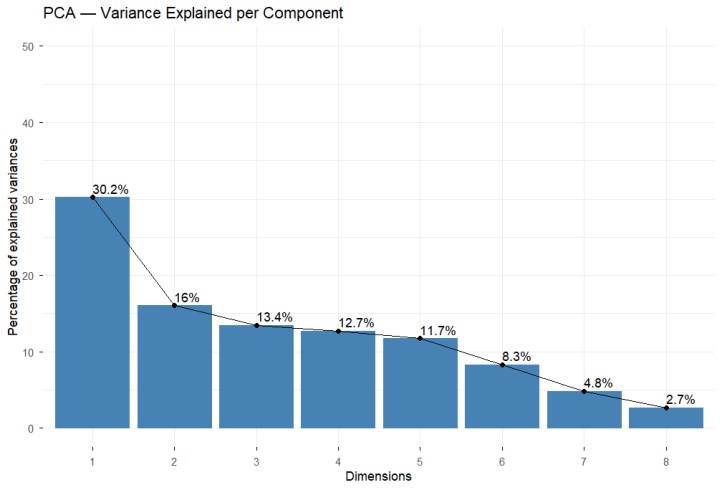


Figure: PCA Variance Explained per Component (Scree Plot)

Variable contribution analysis reveals two interpretable axes. PC1 (Wellness Risk Axis) is dominated by stress_level_0_10 (30%), sleep_hours (26%), and sleep_quality_binary (24%). PC2 (Screen Behaviour Axis) is dominated by work_screen_hours (56%) and leisure_screen_hours (34%). Age, exercise, and social hours contribute little to both; PC3 mainly reflects social hours (44%) and age (29%).

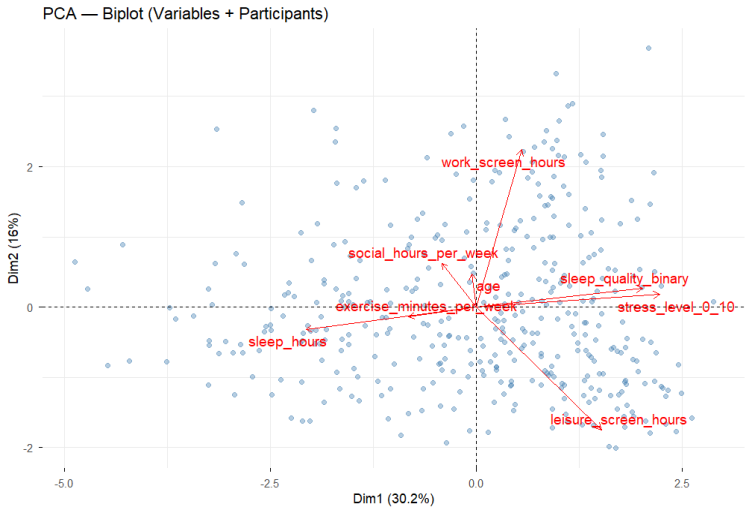


Figure : PCA Biplot (Variables + Participants)

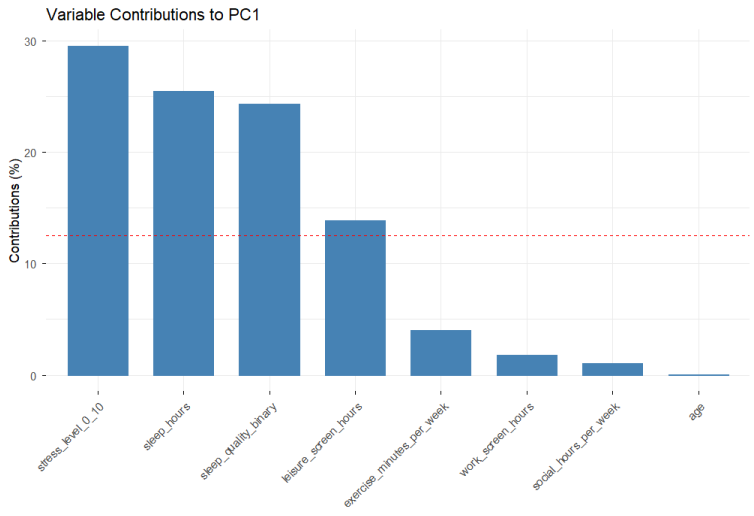


Figure: Variable Contributions to PC1

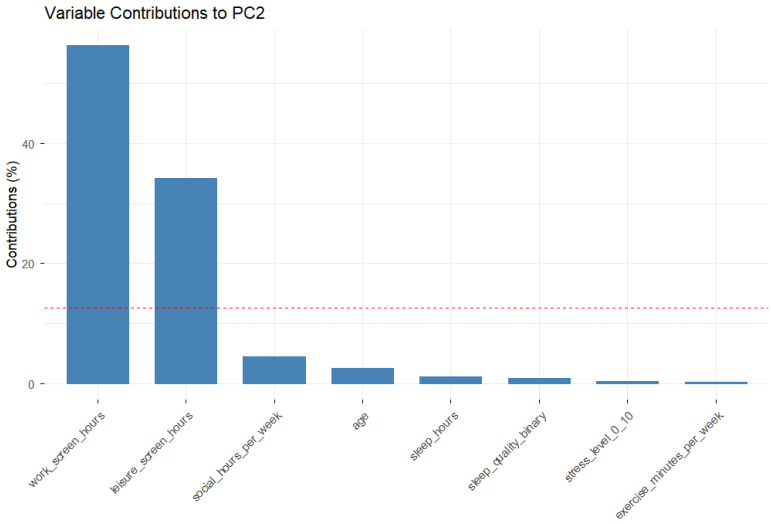


Figure: Variable Contributions to PC2

6.2 Determining the Optimal K

The hierarchical silhouette scores peak at K = 3 (mean silhouette = 0.294) with a sharp drop at K = 4 and no recovery thereafter. In the dendrogram, the last two merges occur at heights 34.4 and 24.3, after which merge heights flatten around 17: there are clear jumps at K = 2 and K = 3 but none beyond. For K-Means, the silhouette is highest at K = 2 (0.349) but almost tied at K = 3 (0.340), and the elbow plot bends gradually rather than sharply. **K = 3 was adopted** because it is

supported by the hierarchical silhouette and separates two high-stress groups that K = 2 merges. All silhouette values are below 0.5, indicating weak-to-moderate cluster structure.

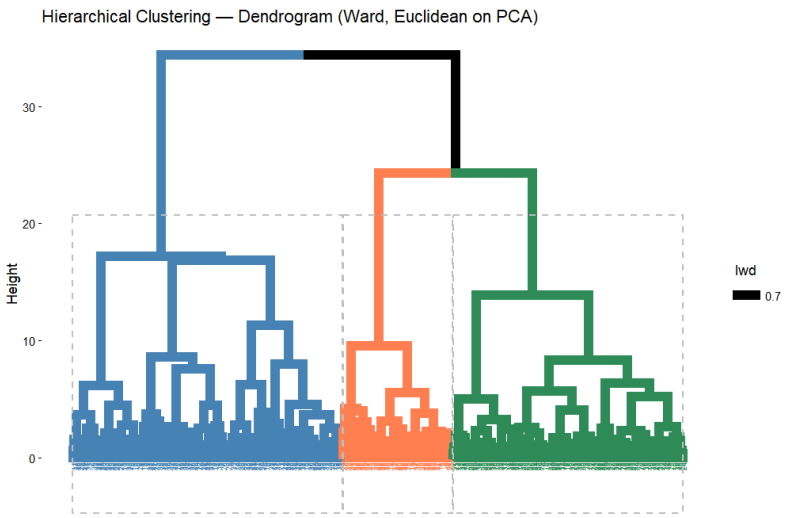


Figure: Hierarchical Clustering Dendrogram (Ward, Euclidean on PCA)

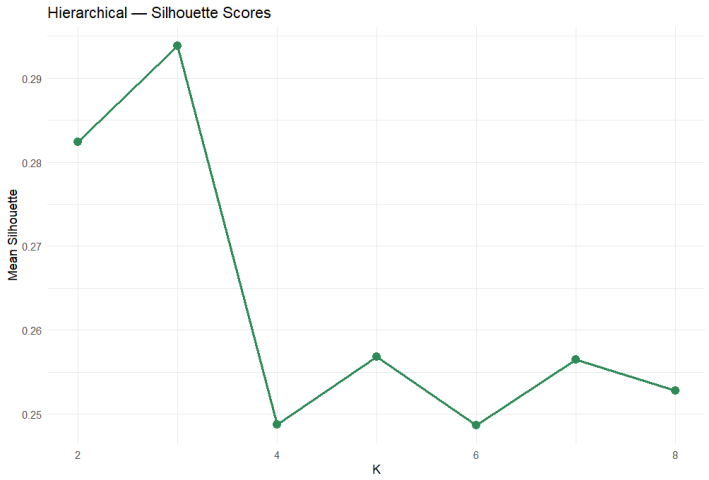


Figure: Hierarchical — Silhouette Scores (K = 2 to 8)

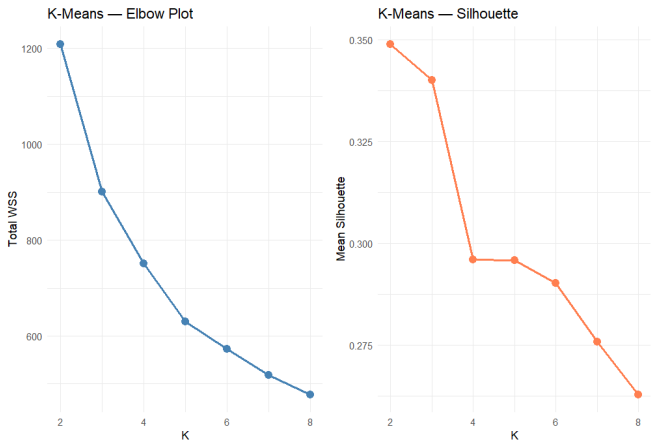


Figure: K-Means — Elbow Plot and Silhouette Scores

6.3 Cluster Profiles

	Cluster 1 Work-screen stressed	Cluster 2 Balanced	Cluster 3 Leisure-screen stressed
N (share)	72 (18%)	177 (44%)	151 (38%)
Mean wellness (0–100)	6.2	35.7	9.0
Mean stress (0–10)	9.57	6.70	9.17
Sleep (hours)	6.62	7.60	6.51
Poor sleep quality (%)	91.7	34.5	98.0
Work screen (h/day)	5.17	1.74	1.28
Leisure screen (h/day)	5.82	5.84	8.51
Exercise (min/week)	93.2	130.9	93.0
Remote workers (%)	98.6	35.0	11.3

Table: Cluster Profiles (Hierarchical, K = 3)

Cluster 1 – Work-screen stressed (N = 72, 18%): Highest stress (9.57), the most work screen time (5.17 h/day) and the lowest wellness (6.2). Almost all members work remotely (98.6%) and half are students, so their screen time is largely tied to remote work or study.

Cluster 2 – Balanced (N = 177, 44%): Lower stress (6.70), longer sleep (7.6 h), more exercise (131 min/week) and far fewer poor sleepers (34.5% vs over 90% in the other clusters). Mean wellness is 35.7, four to six times that of the other two groups. Screen use is moderate.

Cluster 3 – Leisure-screen stressed (N = 151, 38%): High stress (9.17), low wellness (9.0) and the heaviest leisure screen use (8.51 h/day) despite little work screen time (1.28 h/day). Mostly hybrid or in-person workers (11% remote). The pattern is consistent with heavy passive screen consumption, although cross-sectional data cannot show which comes first.

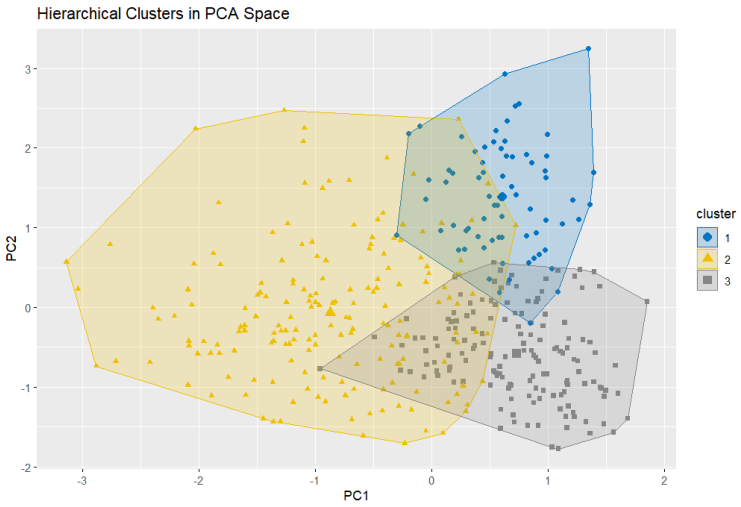


Figure: Hierarchical Clusters in PCA Space (K=3)

While the three clusters are generally distinguishable, they overlap in the middle region of the plot, indicating that not all observations can be cleanly separated.

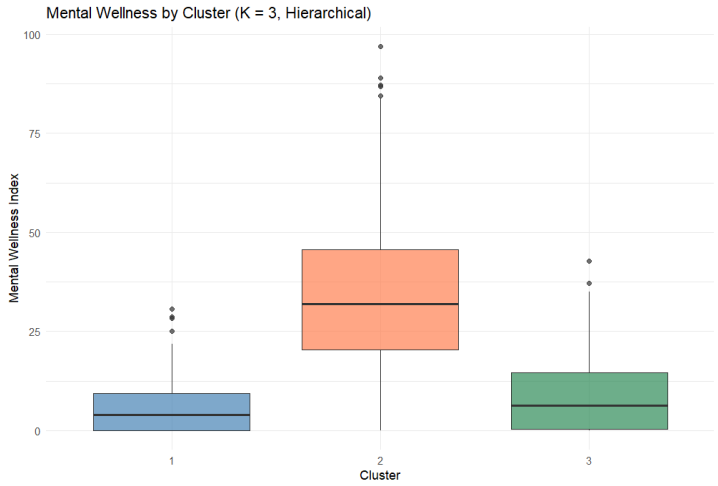


Figure: Mental Wellness Distribution by Cluster (K=3, Hierarchical)

6.4 Cross-Method Validation

To check that the profiles are not an artefact of one algorithm, K-Means was re-run with K = 3 on the same three principal components. Because cluster numbers are arbitrary, labels were matched with the Hungarian algorithm (clue::solve_LSAP) before comparing assignments: the two methods agree on 84.5% of participants. K-Means reaches a slightly higher mean silhouette

(0.340 vs 0.294), and most disagreements (56 of 62) concern participants that the hierarchical method places in the Balanced cluster, which borders both high-stress groups.

The clusters were also compared on variables that were not used to build them. Mental wellness differs strongly between clusters (Kruskal–Wallis $\chi^2 = 204.0$, df = 2, $p < 0.001$), and so does work mode ($\chi^2 = 159.6$, df = 4, $p < 0.001$): 98.6% of Cluster 1 work remotely, against 35.0% of Cluster 2 and 11.3% of Cluster 3.

7. Conclusions

This analysis reveals that participants cluster into three behavioural profiles. Stress and sleep are the primary axes of differentiation, while screen type (work vs leisure) distinguishes two high-risk groups.

Total screen time alone is too coarse a target: the two low-wellness clusters both have high screen time (11.0 and 9.8 h/day) but different compositions, work-driven in Cluster 1 and leisure-driven in Cluster 3. This suggests that different interventions (work–life boundaries vs leisure screen habits) may be relevant, a hypothesis that would need longitudinal data to test.

Limitations: silhouette scores of 0.29–0.34 indicate overlapping rather than sharply separated groups; results depend on retaining three principal components and on K = 3; and the data are self-reported and cross-sectional.